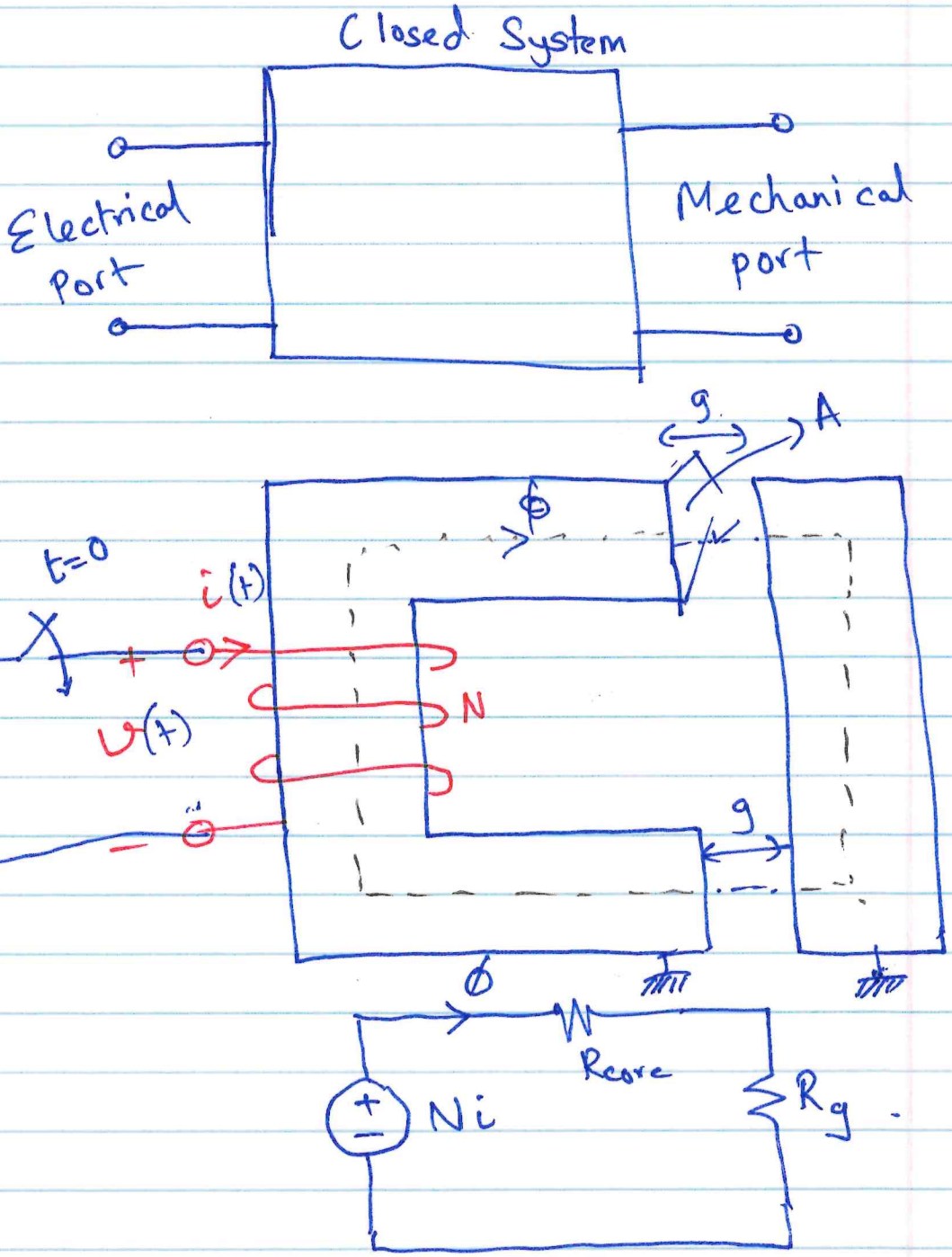
Lecture 13

Electromechanical Energy Conversion.



$$p(t) = v(t)\, i(t)$$

$$E(T) = \int_0^T v(t) \cdot i(t)\, dt \qquad v(t) = L\frac{di}{dt}$$

$$= \int_0^T \left(L\frac{di}{dt}\right) i \, dt$$

$$i(T) = I$$

$$E(T) = L\int_0^I i \, di$$

$$= \frac{1}{2} L I^2$$

$$= \frac{1}{2} I^2 \left[\frac{N^2}{R_c + R_g}\right]$$

$$= \frac{1}{2} \frac{\left[\phi^2 (R_c + R_g)^2\right]}{(R_c + R_g)}$$

$$= \frac{1}{2} \phi^2 \left[R_c + R_g\right]$$

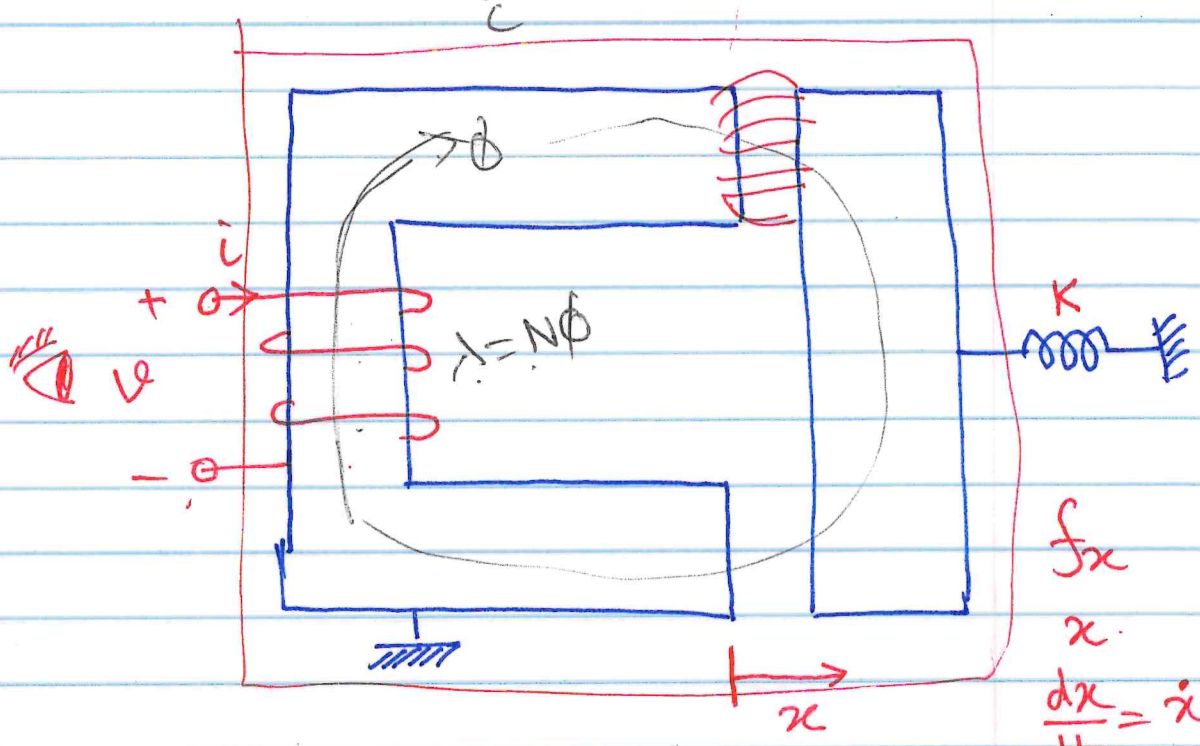
$$= \frac{1}{2}\phi^2 \left[\frac{l_c}{\mu_0 \mu_r A} + \frac{2g}{\mu_0 A} \right]$$

$$= \frac{1}{2}\frac{\phi^2}{A^2 \mu_0} \left[\frac{A \cdot l_c}{\mu_r} + \frac{2Ag}{1} \right]$$

$$= \frac{1}{2}\frac{B^2}{\mu_0} \left[\left[\frac{V_c}{\mu_r} \right] + \left(V_g \right) \right]$$

majority of energy is stored in the gap.

$$\oint_C \underline{F} \cdot d\underline{l} = -\frac{\partial B}{\partial t}$$

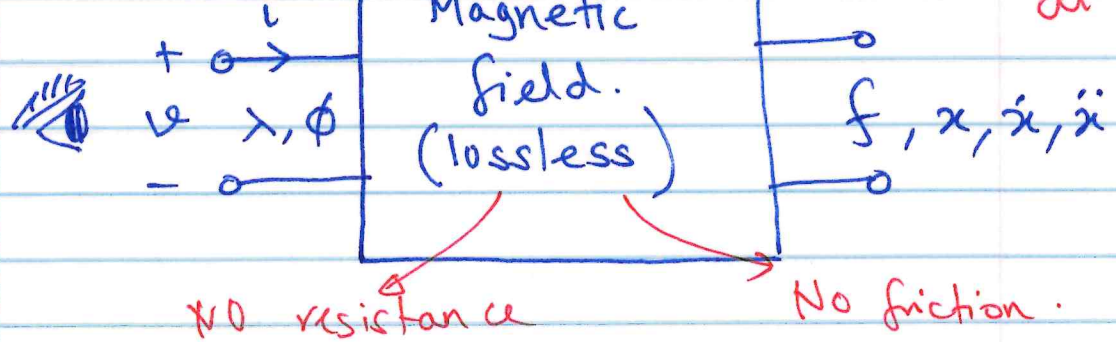


f_x

x

$\frac{dx}{dt} = \dot{x}$

$\frac{d^2x}{dt^2} = \ddot{x}$



Can we come up with an electrical circuit model?

$$\boxed{v = \frac{d\lambda}{dt}} = \lambda = Li$$

$$v = \frac{d\lambda}{dt} = \frac{d}{dt}(Li) = L\frac{di}{dt} + \cancel{i\frac{dL}{dt}}$$

$$= \frac{d[L(x)\,i]}{dt}$$

$$= L(x)\frac{di}{dt} + i\frac{d[L(x)]}{dt}$$

$$= L(x)\frac{di}{dt} + i\frac{\partial L(x)}{\partial x}\cdot\left(\frac{dx}{dt}\right)$$

$$v = \boxed{L(x)\frac{di}{dt}} + \boxed{i\frac{\partial L(x)}{\partial x}\cdot\dot{x}}$$

Speed voltage

back emf

